

KANWAR ROBINSON

Software Development Engineer - II

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Summary

Software engineer who builds AI systems that actually ship. From a customer-facing multi-agent chatbot serving real users to an MCP server now being published across Anthropic's Marketplace, Claude Desktop, and Cursor. Over the past 3+ years at AppViewX, I've cut API response latency by 40%, deflected 35% of support tickets through automation, and architected LLM-driven backends that run in production on Kubernetes at scale. I work across the full stack with LangGraph, RAG, Amazon Bedrock, FastAPI, MongoDB, Kafka. But what drives me is shipping things that work in the real world, not just in demos.

Core Competencies

- **Backend** : Python3.11, FastAPI, Java17, Spring boot 3
- **AI/ML** : LangChain, LangGraph, Retrieval Augmented Generation (RAG), RedisVectorStore
- **DevOps** : Kubernetes, Docker, Helm, Jenkins, GitHub Actions
- **Data & Messaging** : MongoDB, Redis Stack Server, Apache kafka, Zookeeper
- **LLMOps**: Amazon Bedrock, AWS IAM, Cloud Formation, EC2

Professional Experience

AppViewX, Inc.

08/2023 – present | Coimbatore, India

Software Development Engineer - II

- **Architected** and published an **MCP** (Model Context Protocol) server for AppViewX exposing **14 tools** for Certificate Lifecycle Management with OAuth 2.0, enabling users to automate certificate operations via natural language through seamless AppViewX API integration and actively publishing across Anthropic's MCP Marketplace, Claude Desktop, Cursor, and VS Code.
- Led the **end-to-end design** and **development** of a production-grade, customer-facing AI chatbot, owning the **architecture** from day one as the primary individual contributor. Built a foundation with a **LangGraph supervisor architecture** powered by **Amazon Bedrock LLMs**, dynamically routing between specialized sub-graphs and Bedrock driving the supervisor and RBAC sub-agents, while **OpenAI** handles the Documentation assistant. Brought down inbound support tickets by **35%** from appviewx customers.
- Designed and built the **Checkpoint Service** from scratch to handle both **short-term** and **long-term memory** for the chatbot & persisting everything from message history and agent state to tool call results and intermediate steps using LangGraph and Redis Stack Server. Architected it to support **1000 concurrent users** and got it **certified by the PSR team**, making it the backbone of the chatbot's personalization and context continuity across sessions.
- Refactored a latency-critical Python process from a Java-invoked sub-process into a dedicated **FastAPI-based microservice**, deployed within the same **Kubernetes cluster**. This architectural shift reduced API response latency by approximately 40%, significantly **enhancing the chatbot's performance** and scalability.
- Contributed to the development of a **migration framework** for **Redis Vector Store**, enabling seamless transfer of vector embeddings across AppViewX versions to maintain consistency and backward compatibility during feature rollouts.
- Optimized the AppViewX search inventory to manage **20 million records** with **millisecond response** times by implementing advanced query optimization techniques and efficient database design strategies in **MongoDB**.
- Delivered an RBAC (Role-Based Access Control) feature for out-of-the-box solutions, successfully managing multi-project code changes independently under tight deadlines.

AppViewX, Inc.

01/2023 – 07/2023 | Coimbatore, India

Research & Development Intern

- Built an automated **Expiry Alerts** feature for CERT+ using Java Spring Framework as part of a 4-person engineering team, enabling proactive SSL certificate notifications 7-30 days before expiration that reduced manual monitoring workload by 50% and achieved **85% customer adoption within 3 months** and also the **fastest feature adoption** in company history.
- Engineered and deployed a live data streaming solution for transferring millions of documents between **MongoDB** clusters using **Apache Kafka** with **ZooKeeper**, ensuring high availability and zero data loss.

Talakunchi Networks

07/2020 – 08/2020 | Mumbai, India

Cybersecurity & Ethical Hacking Intern

- **Identified and exploited 15+ OWASP Top 10 vulnerabilities** including SQL injection and XSS attacks across 8 production web applications, developed Python scripts to automate server log analysis and anomaly detection, reducing mean time to detection (MTTD) by 40% and contributing to a 60% decrease in open security issues.
- **Conducted penetration testing using Nmap, Burp Suite, and Metasploit** across 5 client environments, mapped attack surfaces, exploited misconfigurations to gain unauthorized access, and documented 25+ security flaws in detailed remediation reports that led to the elimination of 3 critical CVE-rated vulnerabilities within 2 weeks.

Education

Bachelor of Engineering (Mechanical Engineering)

07/2019 – 05/2023

Coimbatore Institute of Technology, Coimbatore, India [GPA - 8.57/10]

Project highlights

Agentic AI Assistant

Built a multi-agent chatbot architecture using **Agentic AI design patterns** with **LangGraph** and **RAG**, enabling dynamic tool use, memory chaining, and goal-driven execution. Built on **FastAPI** and containerized with **Docker**, the system was deployed on **Kubernetes (GKE)** using **Helm** for rollout management, with **OpenTelemetry tracing** and **Prometheus/Grafana** for real-time observability and monitoring.

Kubernetes Runtime Security Management Platform

Built a comprehensive runtime security platform for Kubernetes environments, taking it from architecture to implementation solo. The system continuously monitors live workloads for anomalous behavior, enforces security policies through OPA/Gatekeeper, and detects real-time threats using Falco and all aligned with CIS Kubernetes benchmarks. Built a policy enforcement layer that automatically flags and quarantines non-compliant workloads, and wired up Prometheus and Grafana for a live security observability dashboard. The platform cut down the cluster's vulnerability surface significantly and gave the team proper threat visibility they didn't have before turning what was a blind spot into a fully auditable, policy-driven runtime environment.

Motion Planning and Path Tracing of Robotic Arm using MATLAB and RoboAnalyzer

Implemented a motion planning and path tracing system for a robotic arm using MATLAB and RoboAnalyzer, enabling precise trajectory generation and optimization. The project focused on simulating and analyzing kinematics to enhance robotic movement efficiency.

Laser Powder Bed Fusion of Aluminium Tin Alloy: Manufacturing, Microstructure, and Mechanical Properties

Investigated the manufacturing, microstructure, and mechanical properties of Aluminium-Tin alloy using Laser Powder Bed Fusion and Casting techniques. Conducted tensile, fatigue, and wear tests to optimize composition for enhanced industrial applications.

Awards

Circle of Excellence - AppViewX, Inc.

07/2024

Awarded the **Circle of Excellence 2023** for tackling key challenges in **CERT+**, including expiry alerts, onboarding streamlining, UI enhancements, infrastructure scaling, and driving product-led growth.

Engineer of the Month - AppViewX, Inc.

12/2024

Awarded for implementing **Redis Stack Server** to enhance robust conversational memory and maintain the state in the **LLM engine**.

Spot Award - AppViewX, Inc.

11/2023

Recognized with a Spot Award for efficiently implementing automated Expiry Alert workflows across Jira, Slack, ServiceNow, and CERT+ OOB roles, enhancing workflow automation and operational efficiency.